# SENSITIVITY ANALYSIS

Concerned with how changes in an LP's parameters affect the LP's optimal solution.

. max $Z = 3x_1 + 2x_2$
s.t. $x_1 + 2x_2 \leq 40$
$2x_1 + x_2 \leq 50$
$x_1, x_2 \geq 0$

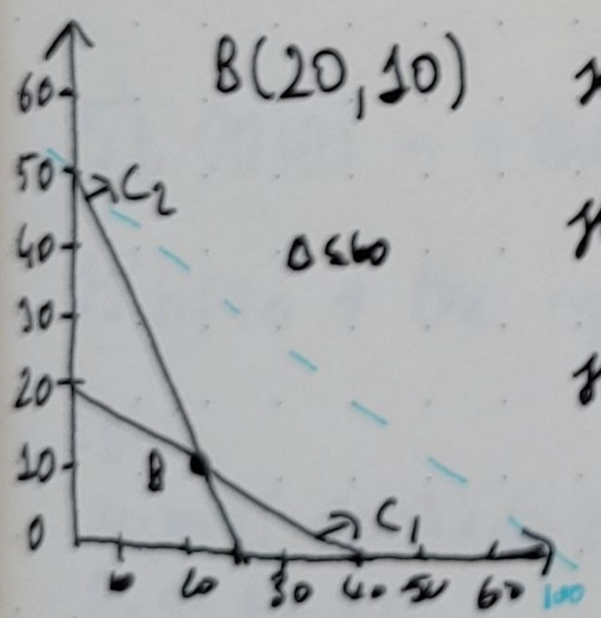
a) For what values of $x_1$ obj. func. coefficient would current basis remain optimal?

b) For what values of $x_2$ obj. Funcion coefficient would current basis remain optimal?

c) If constraint 1 RHS is 30 would the current basis remain optimal? If so what are the values of decision variables?

d) If constraint 2 RHS is 60 would the current basis remain optimal? If so what are the values of decision variables?

e) Find the shadow prices of each constraint.




B(20,10) slope of obj func 3/2

0≤60 slope of $c_1$ 1/2

slope of $c_2$ 2

$\frac{c_1}{2} > 2 \Rightarrow c_1 > 4$, $\frac{c_1}{2} < \frac{1}{2} \Rightarrow c_1 < 1$

$\frac{3}{c_2} > 2 \Rightarrow \frac{3}{2} > c_2$, $\frac{3}{c_2} < \frac{1}{2} \Rightarrow 6 < c_2$

$x_1 + x_2 \leq 100$
$x_1 + x_2 \leq 25$

$x_1 + 2x_2 = 40 + \Delta$
$2x_1 + x_2 = 50$

$2x_1 + 4x_2 = 80 + 2\Delta$
$-2x_1 - x_2 = -50$

$3x_2 = 30 + 2\Delta$
$x_2 = 10 + \frac{2}{3}\Delta$
$x_1 = 20 - \frac{1}{3}\Delta$

$x_1 = 20 - (-\frac{10}{3})$
$x_1 = 20 + \frac{10}{3}$

$x_2 = 10 + \frac{2}{3}(-10)$
$x_2 = \frac{10}{3}$

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# TRANSPORTATION & ASSIGNMENT

## - Vogel's Method

| | D1 | D2 | D3 | D4 | Supply | Row diff → abs(least$_1$ − least$_2$) |
|---|---|---|---|---|---|---|
| O1 | 3 | 1 | 7 | 4 | 300 | 2 |
| O2 | 2 | 6 | 5 | 9 | 400 | 3 |
| O3 | 8 | 3 | 3 | 2 | 500 | 1 |
| Demand | 250 | 350 | 400 | 200 | | |
| Column diff. | 1 | 2 | 2 | 2 | | |

- Find the row or column with the largest penalty and then find the cell with the smallest unit cost in that row/column. Allocate as much as possible to the selected cell, and adjust the associated amounts of supply and demand by subtracting the allocated amount.

- Cross out the row/column with 0 supply/demand. If both arbitrary cross out one.

- If exactly one cell is left uncrossed, cross out the cell and stop. Otherwise, go to step 0.

| | | | | Per |
|---|---|---|---|---|
| 6 | 7 | 8 | $S_1 = 10$ | 1 |
| 15 | 80 | 76 | $S_2 = 15$ | 63 |
| d | 15 | 5 | 3 | |
| Per. | 9 | 73 | 70 | |

⇒

| | | | Sup. |
|---|---|---|---|
| 6 | ~~7~~ | 8 | 5 |
| 15 | ~~80~~ | 76 | 15 |
| Dem | 15 | 5 | |
| | 9 | 70 | |

| | | | Sup. |
|---|---|---|---|
| 06 | ~~7~~ | ~~8~~ | 0 |
| ~~15~~ | ~~80~~ | ~~76~~ | 15 |
| | 15 | | |
| | 9 | | |

$X_{11} = 0$
$X_{12} = 5$
$X_{13} = 5$
$X_{21} = 15$

## - Hungarian Method

| | $J_1$ | $J_2$ | $J_3$ | $J_4$ |
|---|---|---|---|---|
| Crane 1 | 16 | 2 | 3 | 7 |
| Crane 2 | 5 | 8 | 7 | 5 |
| Crane 3 | 8 | 6 | 5 | 9 |
| Crane 4 | 3 | 4 | 5 | 11 |

($n = 4$)

1) Find the min of each row and substract from each row the min value.

2) Find the min of each column and substract from each row the min value.

3) Find the min number of vertical/hr lines to cover all zeroes

If the minimum number of lines is equal to $n$, the optimal sln. is available among the covered zeroes. Otherwise go to step 4

4) Find the min of the uncovered values. Then substract the min from all the uncovered values and add it to the intersection points Then go back to the step 3.

5) Start the assignment from the row/column that has minimum numbe of zeroes Cross allocated row/column. Update number of zeroes.

Step 1 and 2

| | | | |
|---|---|---|---|
| 16 | 2 | 3 | 4 |
| 3 | 0 | 7 | 4 |
| 6 | 1 | 0 | 0 |
| 1 | 0 | 4 | 10 |

$\Rightarrow$

| | | | |
|---|---|---|---|
| 14 | 0 | 1 | 2 |
| 3 | 0 | 4 | 1 |
| 6 | 1 | 0 | 0 |
| 1 | 0 | 4 | 10 |

$\Rightarrow$

| | | | |
|---|---|---|---|
| 13 | 0 | 0 | 1 |
| 2 | 0 | 3 | 0 |
| 5 | 0 | 0 | 2 |
| 0 | 0 | 0 | 6 |

$\Rightarrow$

| | | | |
|---|---|---|---|
| 12 | 0 | 0 | 0 |
| 2 | 0 | 3 | 0 |
| 5 | 0 | 0 | 2 |
| 0 | 0 | 0 | 6 |

$x_{11} = 16$ $x_{34} = 9$

$x_{21} = 3$ $x_{43} = 5$

$3 \neq (n=4)$ $3 \neq (n=4)$ $4 = n$